Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, product preferences, and subscription behavior from **3,900 transactions** to guide strategic business decisions.

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Dataset at a Glance

3,900
Transactions

18
Features

37
Missing Ratings

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping Type

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Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for initial inspection.

02

Clean & Standardize

Imputed missing Review Ratings with category median; renamed columns to snake_case.

03

Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

04

Database Load

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

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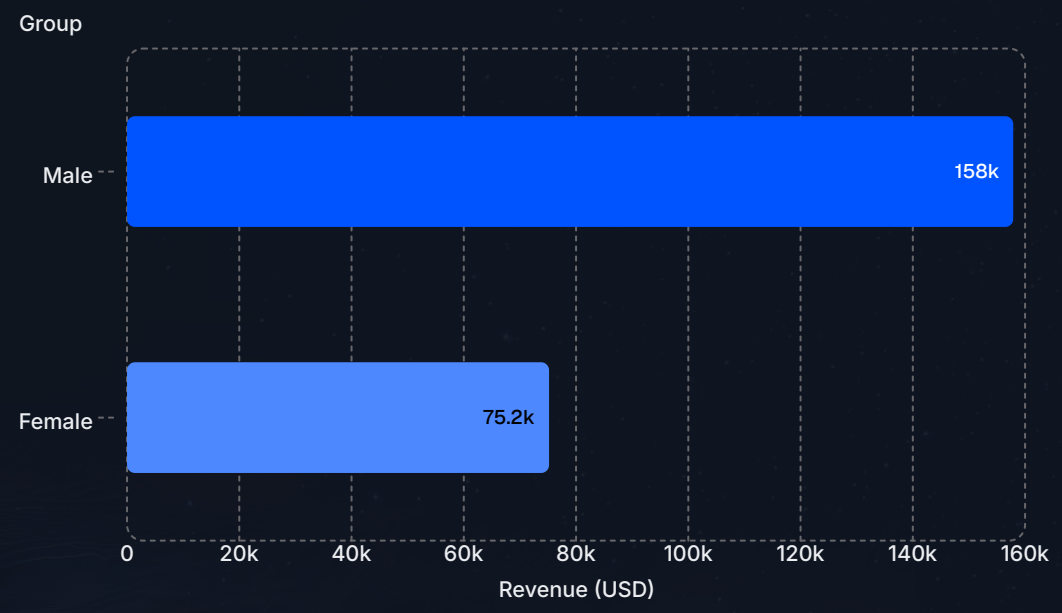
Revenue & Spending Insights



Key Findings

- **Male customers** generated $157,890 vs. **Female** at $75,191 in total revenue.
- **Express shipping** users averaged $60.48 vs. $58.46 for Standard — a small but notable premium.
- **839 customers** used discounts yet still spent above the average purchase amount.

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Top-Rated Products & Discount Dependency

Top 5 Products by Avg. Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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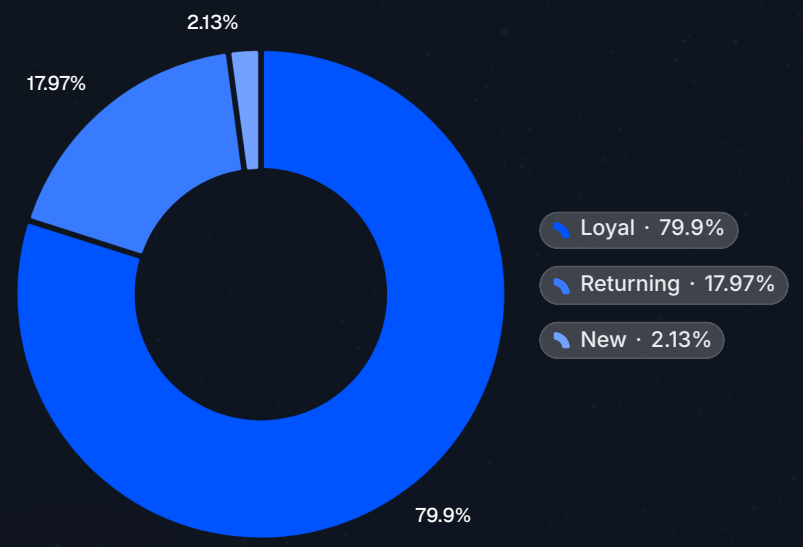
Customer Segmentation



Segment Breakdown

Customers were classified based on purchase history into three segments:

- **Loyal (3,116):** High repeat purchase count — the dominant group.
- **Returning (701):** Moderate engagement with growth potential.
- **New (83):** Recently acquired customers to nurture.

Repeat buyers with >5 purchases: **958 subscribers** vs. **2,518 non-subscribers.**

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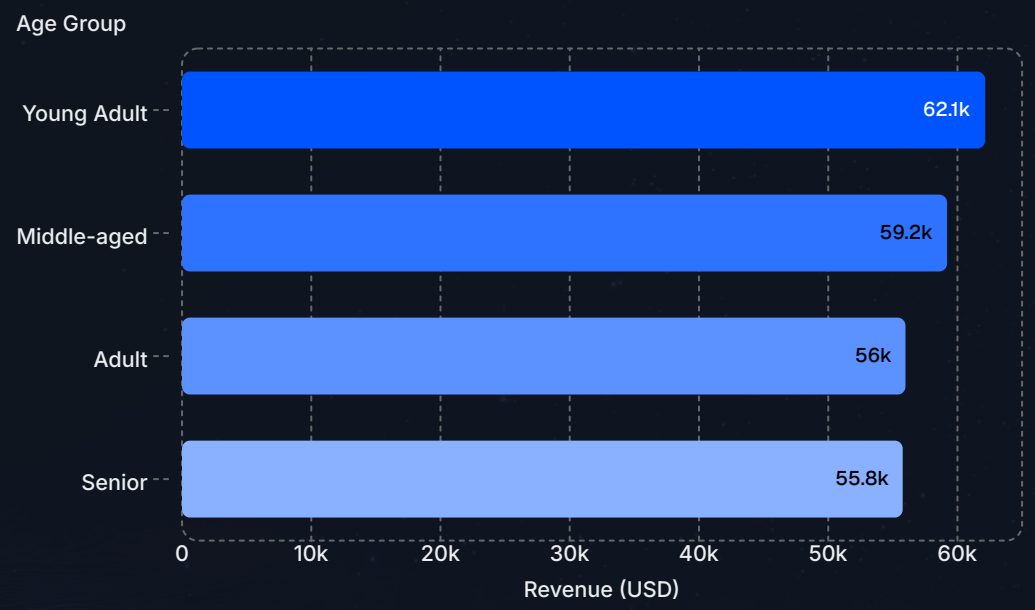
Subscription & Revenue by Age Group



Subscription Status

Status	Customers	Avg. Spend	Total Revenue
Subscribers	1,053	$59.49	$62,645
Non-Subscribers	2,847	$59.87	$170,436

Young Adults lead in revenue. Avg. spend is nearly identical across subscription status — volume drives the gap.

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Top Products by Category

👗 Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

💍 Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

👟 Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

🧥 Outerwear

- Jacket — 163
- Coat — 161

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Power BI Dashboard

3.9K Customers	**$59.76** Avg. Purchase
3.75 Avg. Rating	**27% / 73%** Subscribers vs. Non

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control — especially for Hat, Sneakers, and Coat.

Targeted Marketing

Focus on high-revenue Young Adults and express-shipping users; highlight top-rated products.

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